UNIQUENESS OF THE RADIAL L_1^d EXTREMAL

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> Prove that the radial extremal function in the problem L_1^d is unique not only among radial functions.

Problem 7, printed p. 37, of Dmitry Gorbachev, *Sharp Bernstein–Nikolskii inequalities for polynomials and entire functions of exponential type*, arXiv:2609.32072v1. The class consists of integrable entire functions of spherical exponential type at most one on $\mathbb{R}^d$; the reconstructed question is whether its normalized radial minimizer is unique without the radial restriction.

Answered: yes. Evidence: Sketch.

The normalized radial minimizer is the unique minimizer in the full class for every $d \geq 1$.

Proof. Write φ_d for the radial minimizer, normalized by $\varphi_d(0) = 1$, and $m_d = \|\varphi_d\|_{L^1(\mathbb{R}^d)}$. Given any admissible minimizer f, equality in $\|\operatorname{Re} f\|_1 \leq \|f\|_1$ shows that f is real on $\mathbb{R}^d$. Take its central even part $g(x) = (f(x) + f(-x))/2$. For almost every unit vector ω, the even entire function $t \mapsto g(t\omega)$ has finite radial integral with weight $|t|^{d-1}$. Its even power series defines a radial entire function on $\mathbb{C}^d$ of spherical type at most one. It is therefore an admissible radial competitor. Radial uniqueness and polar integration force equality in the raywise lower bound for almost every ω; hence $g = \varphi_d$. The central averaging inequality then shows that the odd part $h = f - g$ satisfies $|h(x)| \leq |\varphi_d(x)|$ on every real ray.

We use the following division fact. If Φ is even entire of exact type $\sigma > 0$, has only simple real zeros, and is of bounded type in both half-planes, while H is entire of type at most σ, of bounded type in both half-planes, and $|H(x)| \leq |\Phi(x)|$ on the real axis, then $H = c\Phi$. Indeed, every zero cancels in $q = H/\Phi$, so q is entire, of bounded type in both half-planes, and bounded on the real axis. Evenness and exact type give mean type σ for Φ in each half-plane. Additivity of mean type gives nonpositive mean type for q in each half-plane. Krein's theorem then makes q zero type; the Phragmén–Lindelöf principle for a zero-type entire function bounded on the real axis makes it constant; see de Branges, *Hilbert Spaces of Entire Functions*, Theorems 9–10 and Problem 29, https://www.math.purdue.edu/~branges/Hilbert%20Spaces%20of%20Entire%20Functions.pdf.

The radial profile $\Phi(z) = \varphi_d(ze_1)$ has exact type one: otherwise a slight dilation would reduce its L^1 norm while preserving admissibility. Its zeros are simple and real by the radial extremal theorem of Dai, Gorbachev and Tikhonov, *Estimates of the asymptotic Nikolskii constants for spherical polynomials*, Theorem 4.2(i), doi:10.1016/j.jco.2021.101553. The restrictions of f and φ_d to each real line have bounded type in both half-planes, by Fourier inversion for functions with compact spectral support. Apply the division fact to $H(z) = h(z\omega)$ and Φ on each ray. It gives $h(z\omega) = c_\omega \varphi_d(z\omega)$; since h is odd and φ_d even, $c_\omega = 0$. Thus $h = 0$ on all real rays and $f = \varphi_d$. □

AI assistance was used in adapting and checking the argument and the exposition.

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